

**EE-1005: Digital Logic
Design**

BS-CS & BS-SE

Date: 4th April, 2024

Course Instructor(s)

Mr. Shams Farooq, Dr. Mehwish Hassan,
Mr. Muhammad Aamir Gulzar

Sessional-II Exam

Total Time: 1 Hour

Total Marks: 50

Student Name

Roll No.

Course Section

Student Signature

Do not write anything on the question paper except the information required above.

Instructions:

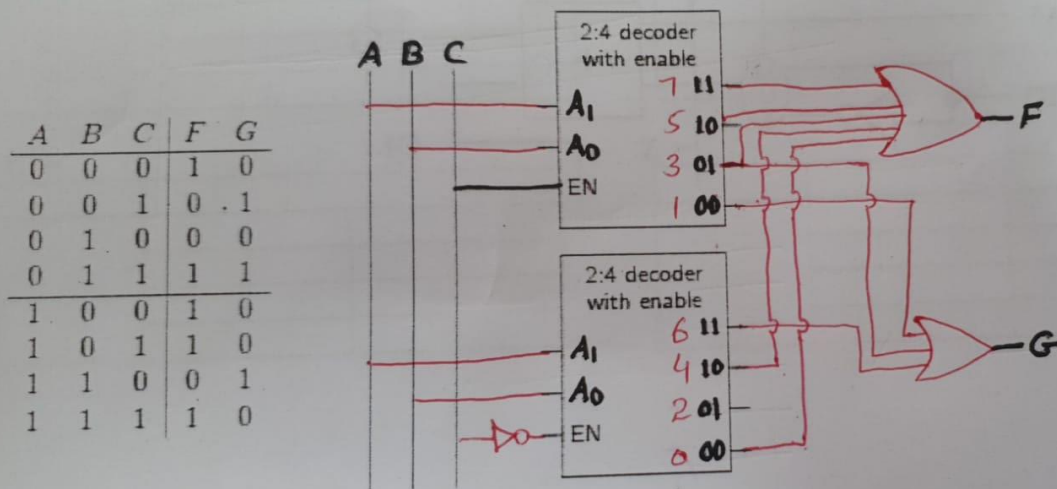
1. Read the question carefully, understand the question, and then attempt your answers in the provided answer booklet.
2. Verify that you have **One? (1?)** printed page of the question paper including this page. There are **Five? (5?)** questions.
3. Calculator sharing is strictly prohibited.

Q1: Question statement

[10 marks]

Shown below are two 2:4 decoders-with-enable. Add wires, one not gate, and two or gates to implement the functions F and G given in the truth table.

Q1: Shown below are two 2:4 decoders-with-enable. Add wires, one not gate, and two or gates to implement the functions F and G given in the truth table. [10]



$$F = \sum(0, 3, 4, 5, 7)$$

$$G = \sum(1, 3, 6)$$

Q2: Question statement

[10 marks]

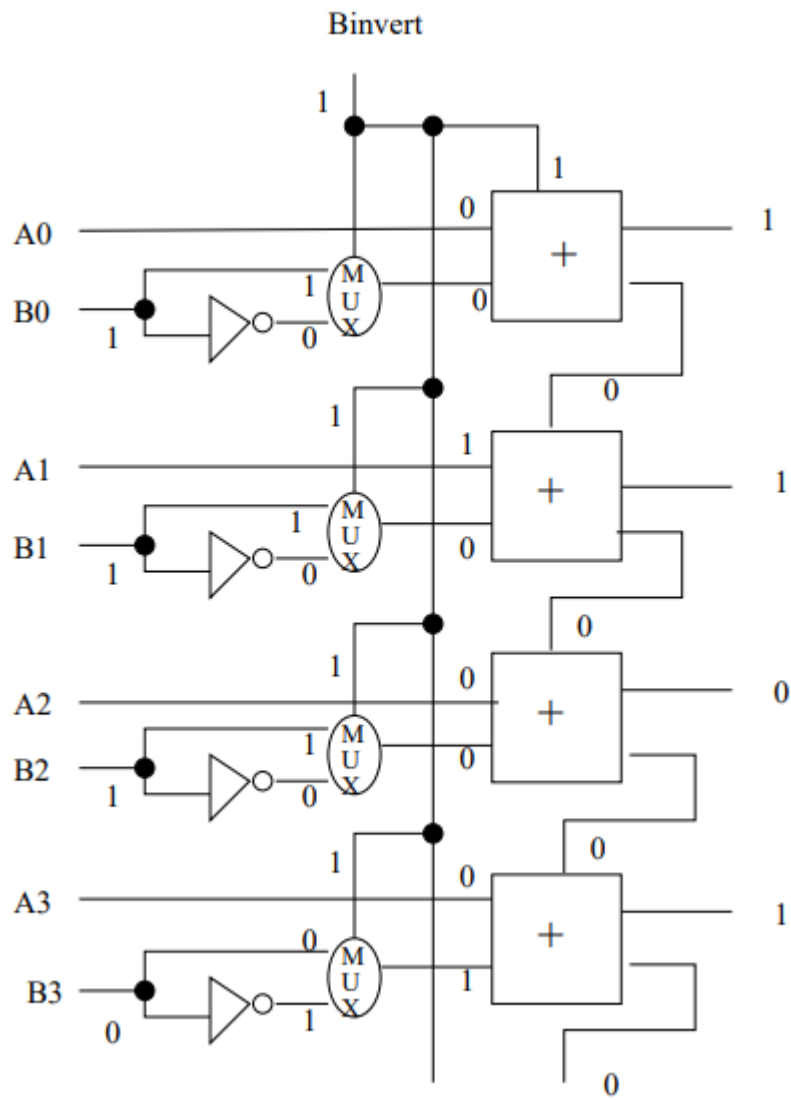
Consider the low order four bits of the adder/subtractor circuit shown below. Label all wires (inputs, outputs) in the circuit by their value (0 or 1), assuming the circuit is computing $A - B$ where $A = 98$ and $B = -25$. [Do not draw the entire circuit. Only consider the four lowest order bits for this question.]

$A = 98$ in binary (signed) is ...0001100010, so four low order bits of A are 0010

$B = -25$ in binary (signed) is ...1111100111, so four low order bits of A are 0111

$A - B = 98 - (-25) = 123 = (001111011)_2$, so four low order bits are 1011.

Verify that this is consistent with the 0's and 1's on the wires (which you can compute independently by examining the sum and carry bits).



Q3: Question statement

[10 marks]

Design a four-bit combinational circuit 2's complemener. (The output generates the 2's complement of the input binary number.) Show that the circuit can be constructed with exclusive-OR gates.

Truth Table

I3	I2	I1	I0	O3	O2	O1	O0
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0
0	0	1	1	1	1	0	1
0	1	0	0	1	1	0	0
0	1	0	1	1	0	1	1
0	1	1	0	1	0	1	0
0	1	1	1	1	0	0	1
1	0	0	0	1	0	0	0
1	0	0	1	0	1	1	1
1	0	1	0	0	1	1	0
1	0	1	1	0	1	0	1
1	1	0	0	0	1	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	0	1	0
1	1	1	1	0	0	0	1

KMaps:

O3(I3,I2,I1,I0) = ?

I3I2\I1I0	I1'I0'	I1'I0	I1I0	I1I0'
I3'I2'	0	1	1	1
I3'I2	1	1	1	1
I3I2'	0	0	0	0
I3I2	1	0	0	0

O2(I3,I2,I1,I0) = ?

I3I2\I1I0	I1'I0'	I1'I0	I1I0	I1I0'
I3'I2'	0	1	1	1
I3'I2	1	0	0	0
I3I2'	1	0	0	0
I3I2	0	1	1	1

O1(I3,I2,I1,I0) = ?

I3I2\I1I0	I1'I0'	I1'I0	I1I0	I1I0'
I3'I2'	0	1	0	1
I3'I2	0	1	0	1
I3I2'	0	1	0	1
I3I2	0	1	0	1

O0(I3,I2,I1,I0) = ?

I3I2\I1I0	I1'I0'	I1'I0	I1I0	I1I0'
I3'I2'	0	1	1	0
I3'I2	0	1	1	0
I3I2'	0	1	1	0
I3I2	0	1	1	0

Q3: Design a four-bit combinational circuit 2's complementor. (The output generates the 2's complement of the input binary number.) Show that the circuit can be constructed with exclusive-OR gates [10]

Truth Table

D	C	B	A	W	X	Y	Z
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0
0	0	1	1	1	1	0	1
0	1	0	0	1	1	0	0
0	1	0	1	1	0	1	1
0	1	1	0	1	0	1	0
0	1	1	1	1	0	0	1
1	0	0	0	1	0	0	0
1	0	0	1	0	1	1	1
1	0	1	0	0	1	1	0
1	0	1	1	0	1	0	1
1	1	0	0	0	1	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	0	1	0
1	1	1	1	0	0	0	1

K-Maps:

$W = D'C + D'B + D'A + DCBA'$

DC/BA	00	01	11	10
00	0	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$X = C'B + CA + BC'A'$

CB/A	00	01	11	10
00	0	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$Y = BA' + A'B$

BA	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	1	0	0	1
10	1	0	0	1

$Z = A$

A	0	1
00	0	1
01	0	1
11	1	1
10	1	1

$$W = D \oplus (C + B + A)$$

$$Y = C \oplus (B + A)$$

$$Y = B \oplus A$$

$$Z = A$$

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Q4: Question statement

[10 marks]

An 8 x 1 multiplexer has inputs A, B, and C connected to the selection inputs S_2 , S_1 , and S_0 , respectively. The data inputs I_0 through I_7 are as follows:

(a) $I_1 = I_2 = I_7 = 0$; $I_3 = I_5 = 1$; $I_0 = I_4 = D$; and $I_6 = D'$.

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Q4: An 8x1 multiplexer has inputs A, B, and C connected to the selection inputs S_2 , S_1 , and S_0 , respectively. The data inputs I_0 through I_7 are as follows:
(a) $I_1 = I_2 = I_7 = 0$; $I_3 = I_5 = 1$; $I_0 = I_4 = D$; and $I_6 = D'$. [10]

A	B	C	D	F^{16}	F^8	
0	0	0	0	0	D	I_0
0	0	0	1	1		
0	0	1	0	0	0	I_1
0	0	1	1	0		
0	1	0	0	0	0	I_2
0	1	0	1	0		
0	1	1	0	1	1	I_3
0	1	1	1	1		
1	0	0	0	0	D	I_4
1	0	0	1	1		
1	0	1	0	1	1	I_5
1	0	1	1	1		
1	1	0	0	1	$\bar{D}$	I_6
1	1	0	1	0		
1	1	1	0	0	0	I_7
1	1	1	1	0		

$(A, B, C, D) = \Sigma(1, 6, 7, 9, 10, 11, 12)$

Q5: Question statement

[? marks]

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Q5: Simplify the following Function using K-Map and implement it with two level NOR gate circuit. [10]

$$\bar{A}C\bar{D} + \bar{A}BD + ABD + (\bar{A} + B + \bar{C} + \bar{D}) + (\bar{A} + \bar{B} + \bar{D})$$

$$\bar{A}C\bar{D} + \bar{A}BD + ABD + \bar{A}\bar{B}CD + \bar{A}\bar{B}D$$

$$\begin{array}{c} \bar{A}B\bar{C}\bar{D} \\ 0010 \\ \hline 0110 \\ \hline m_6 \end{array} + \begin{array}{c} \bar{A}B\bar{C}D \\ 0101 \\ \hline 0111 \\ \hline m_7 \end{array} + \begin{array}{c} \bar{A}BCD \\ 1101 \\ \hline 1111 \\ \hline m_{15} \end{array} + \begin{array}{c} A\bar{B}CD \\ 1011 \\ \hline 1011 \\ \hline m_{11} \end{array}$$

AB \ CD	00	01	11	10
00	0	0	0	1
01	0	1	1	1
11	0	1	1	0
10	0	0	1	0

$(C+D) (C+B) (\bar{A}+D) (A+B+\bar{D})$

$\equiv$